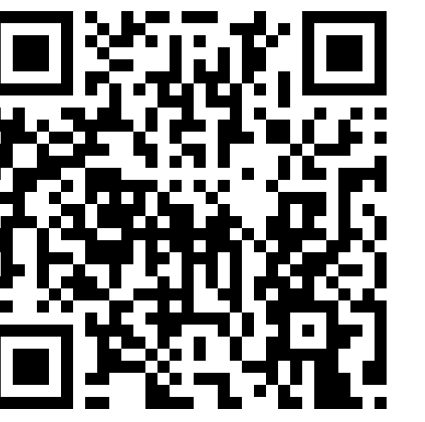




1 The problem

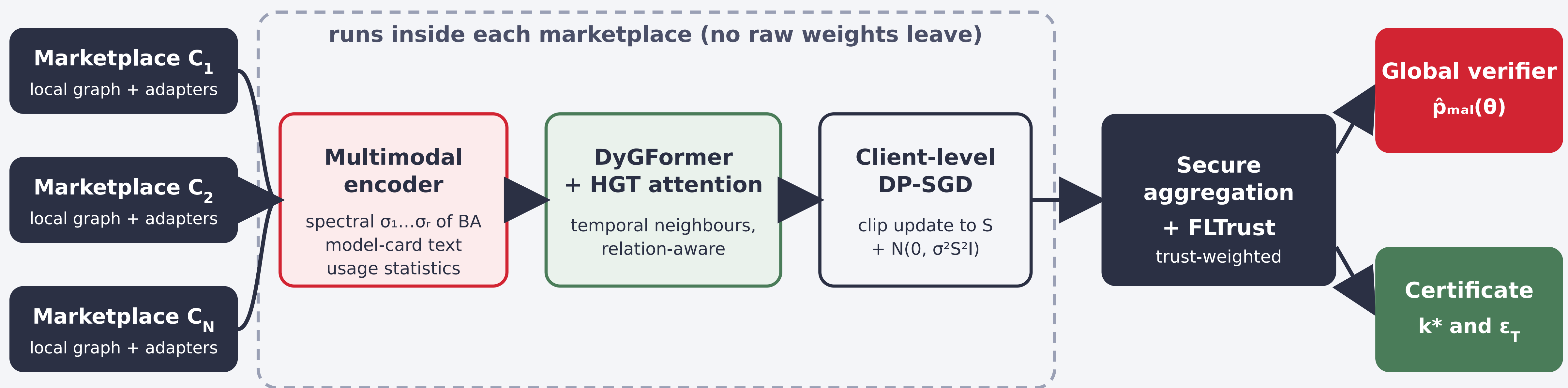
- LoRA is the dominant way to fine-tune LLMs, and public hubs now host **tens of thousands** of shared adapters.
- Backdoored adapters **merge silently** into task adapters and keep their benign utility; a cheap LoRA fine-tune can strip a chat model's safety alignment.
- Centralised detectors (e.g. PEFTGuard) need marketplaces to **hand over raw weights**. That conflicts with proprietary practice, creates a single point of trust, and misses **cross-marketplace lineage**.

2 Our idea

Model the whole adapter ecosystem as a **dynamic graph**, learn a detector **federatedly** under **differential privacy**, and **certify** how many colluding marketplaces it can withstand.

No marketplace ever shares raw adapter weights; only clipped, noised, securely aggregated updates leave a client.

3 Framework



The ecosystem is a **heterogeneous continuous-time dynamic graph**: **adapters** **base models** **contributors** **applications** linked by time-stamped **contributes** **derives_from** **fine_tunes** **deploys** **cites** **co_uses** edges.

96.4%macro-F1 on
LoRAchain-2026**25/50**certified coalition bound k^*
($\epsilon_T=0.5$, bounded-update)**1.7 pp**from centralised PEFTGuard,
with no weight sharing **≈ 13.6** training-time ϵ_T ($\delta=10^{-5}$)
Opacus PRV, $T=100$ **0.7 s**per adapter on A100
(1.6 s A10G · 2.9 s T4)

4 Training round

- Server Poisson-samples marketplaces at rate $q = 0.2$ (10 of $N = 50$).
- Each client computes its update, **clips the whole update** to $S = 1.0$ and adds one Gaussian draw ($\sigma = 1.1$).
- Updates travel through **secure aggregation**; **FLTrust** weights each by $\text{ReLU}(\cos)$ to a vetted 200-adapter root set.
- An **Opacus PRV accountant** tracks the cumulative privacy budget: $\epsilon_T \approx 13.6$ at $T = 100$.
- Measured cost of DP + SecAgg: **$\approx 4\%$** extra wall-clock per round vs. non-private FedAvg.

5 Theory

Thm 1 (client-update sensitivity). With GELU ($\rho \approx 1.13$) and relation weights $\|W_R\| \leq B_W$:

$$\Delta_2 \leq \min(2S, 2S \cdot (\rho B_W d_{\max})^{L \cdot |R|^{1/2}})$$

Thm 2 (certified poisoning radius). Under a bounded-update attack, the verdict on an adapter is invariant w.p. $\geq 1 - T\delta$ for any coalition of size

$$k \leq k^* = \lfloor N(\Phi^{-1}(\hat{p}_1) - \Phi^{-1}(\hat{p}_2)) / (2e^{\epsilon_r}) \rfloor$$

With $N = 50$, $\hat{p}_1 - \hat{p}_2 \approx 0.6$, $\epsilon_r = 0.5$: **$k^* = 25$** . The certificate uses the per-query smoothing budget ϵ_r , not the cumulative ϵ_T .

Certified bound (50%, bounded-update threat) and empirical label-flip tolerance ($\rho \leq 0.33$) are distinct results.

6 Mechanism: the spectral signature

Backdoored adapters concentrate trigger information in one direction: the **top singular value of BA is 2.43 $\times$ larger** than for benign adapters, while trailing values are indistinguishable (KS test $D = 0.71$, $p < 10^{-3}$). The weight modality extracts this; the graph propagates it along lineage.

A **spectral-only (SVD) baseline** reaches 90.2% on simple rank-1 backdoors but only **71.4%** on composite, weight-poisoning and trigger-free merging attacks, and gives no certificate.

7 Main results

Method	ACC	Macro-F1	AUROC	k^*	ϵ_T
PEFTGuard (centralised)	98.3	98.1	0.991	N/A	N/A
ShadowGenes (centralised)	95.1	94.7	0.967	N/A	N/A
FedAvg-MLP	89.6	88.4	0.924	0	N/A
DP-FedAvg-MLP	86.8	85.7	0.901	13	13.6
FedGraphNN-HGT	92.7	91.6	0.951	0	N/A
Krum-DyGFormer	93.5	92.8	0.962	6	N/A
FedLoRAGuard (ours)	97.1	96.4	0.984	25	13.6

LoRAchain-2026: 13,500 benign + backdoored adapters · 4 base-model families (Llama-2/3, Mistral, Qwen) · 10 attack types · ranks 4–32. Mean of 3 seeds; Friedman $\chi^2_F(6) = 51.4$, $p < 10^{-8}$.

8 Ablation: what matters

Configuration	Macro-F1	k^*
Full FedLoRAGuard	96.4	25
w/o DyGFormer (static GAT)	92.0	16
w/o HGT attention	93.4	19
w/o multimodal (weights only)	94.7	22
w/o FLTrust	95.6	19
w/o DP-SGD	96.9	0

Temporal lineage (DyGFormer) carries the most signal. Removing DP gains only 0.5 pp but **destroys the certificate**: the privacy cost buys the guarantee.

9 Robustness

- Stricter splits**: lineage-disjoint 95.1 · base-model-disjoint 93.7 · attack-family held-out 92.3 (vs. 96.4 random split).
- Missing lineage edges**: 95.6 / 93.8 / 91.1 at 10 / 30 / 50% removed.
- Stronger privacy**: 94.1% macro-F1 at $\epsilon_T = 1.0$.
- Byzantine label-flip**: no certificate loss up to $\rho \leq 0.33$.
- Transfer**: same backbone reaches 95.8% macro-F1 on 4 network-IDS datasets.

10 Take-aways

- Adapter integrity can be verified **across marketplaces without sharing weights**, within 1.7 pp of a centralised detector.
- Lineage over time** is the key signal beyond each adapter's own weights.
- DP turns into a **verifiable guarantee**: a certified bound on colluding marketplaces.

11 Scope & limits

- Certificate covers **bounded-update** adversaries, not arbitrary Byzantine gradients.
- Lineage edges are partly **synthesised** from HuggingGraph and PADBench.
- 50/50 benchmark split is a design choice; deploy with a low-FPR threshold and human review.

